

Fortinet Report

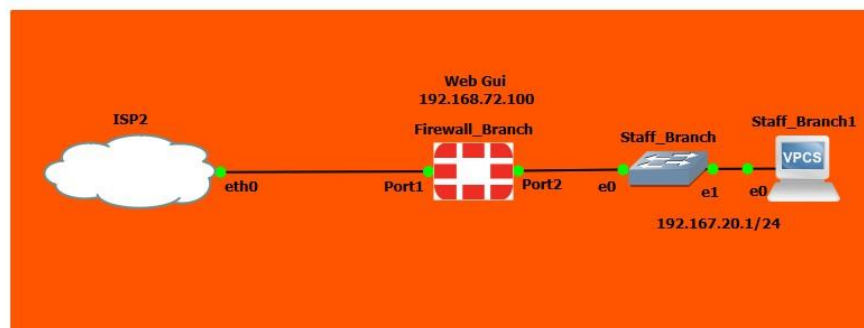
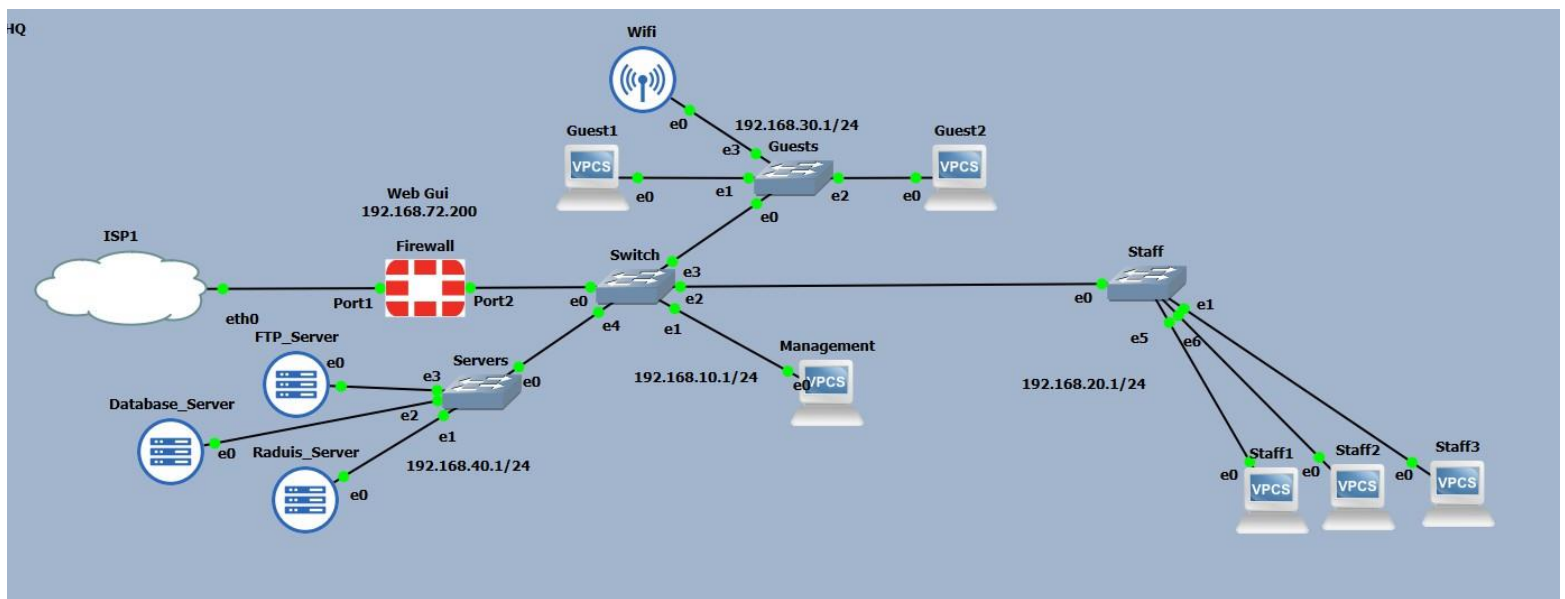
Network Security Fundamentals and FortiGate Integration

#2 adding

/By

Sara Hatem Mohamed	21043483
Rawda Ayman Saad	21032392
Mostafa Hossam eldein Ebrahim	21001727
Shaza Muhmad Sofy Abdelwahed	21055286
Omar Adil Selim	21043962

The network topology



Policies after

+ Create New Edit Delete Policy Lookup Search											Interface Pair View	By Sequer
ID	Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes		
HQ-to-Branch → Staff (Vlan 20)												
5	vpn_HQ-to-Branch_remote	HQ-to-Branch_remote	HQ-to-Branch_local	always	ALL	ACCEPT	Disabled	default certificate-inspection	All	4.90 kB		
Servers (Vlan 40) → Staff (Vlan 20)												
7	Guests Internet	all	all	always	DNS FTP MYSQL	ACCEPT	Enabled	default default default default	All	0 B		
Staff (Vlan 20) → HQ-to-Branch												
4	vpn_HQ-to-Branch_local	HQ-to-Branch_local	HQ-to-Branch_remote	always	ALL	ACCEPT	Disabled	default certificate-inspection	All	0 B		
Staff (Vlan 20) → Servers (Vlan 40)												
6	Staff To Servers	all	all	always	FTP MYSQL RADIUS	ACCEPT	Enabled	default certificate-inspection	All	0 B		
Staff (Vlan 20) → WAN (port1)												
3	Staff Internet	all	all	always	ALL	ACCEPT	Enabled	default default default default	All	216 B		
Implicit												
0	Implicit Deny	all	all	always	ALL	DENY			Disabled	2.52 kB		

ID 3: Staff Internet

Purpose:

Allow staff PCs (Vlan 20) to browse the internet via the WAN interface.

Explanation:

This is a standard outbound internet access rule. All services are allowed (HTTP, HTTPS, DNS, etc.), and NAT is enabled so staff devices can use the firewall's public IP. Security profiles are applied to protect the users from malicious sites and traffic.

Why it's important:

This is the primary user internet access policy, essential for real-world network functionality.

ID 4: pn_HQ-to-Branch_local

Purpose:

This is the **return traffic rule** for the VPN — enabling the Branch Office to communicate back to the HQ internal LAN.

Explanation:

Traffic coming **from the remote branch** (vpn_HQ-to-Branch_remote) is allowed into the HQ local network (HQ-to-Branch_local). This policy is symmetrical with Policy 5, enabling **two-way communication** between HQ and Branch via VPN. NAT is disabled because VPN communication must use real internal IPs, not translated ones.

Why it's important:

Without this policy, the VPN tunnel would be **one-way only**, breaking branch-to-HQ connectivity.

ID5: HQ-to-Branch

This policy enables secure communication between the **Headquarters LAN (Staff Vlan 20)** and the **Branch Office network** through a VPN tunnel.

Explanation:

Traffic originating from the Staff VLAN (192.168.20.0/24) is allowed to pass through the VPN tunnel interface (vpn_HQ-to-Branch_remote) toward the branch network (HQ-to-Branch_local). The service is set to **ALL**, meaning any protocol is permitted. NAT is disabled because VPN tunnels require **routing, not NAT**, to maintain end-to-end addressing. Security profiles (“default, certificate-inspection”) ensure encrypted traffic is validated.

Why it’s important:

This creates the **site-to-site connectivity** needed for HQ employees to access branch resources securely.

ID6: Staff To Servers

Purpose:

Allows devices in Staff VLAN (20) to access required services on Servers VLAN (40).

Explanation:

This rule permits staff to reach server resources using only the protocols needed:

FTP – File transfers.

MySQL – Database access

RADIUS – Authentication

NAT is enabled for traffic translation between VLANs. Basic security is applied using Antivirus and SSL Inspection, while other filters are disabled since the traffic is internal. Logging is enabled for all sessions.

Why it’s important:

Ensure staff can access essential server services while limiting unnecessary or unauthorized traffic, improving both security and control.

Name	0	Staff_to_Servers
Incoming Interface	ell.	Staff (Vian 20) ...
Outgoing Interface	ell.	Servers (Vian 40) ...
Source	☐ all	X
		+
Destination	☐ all	X
		+
Schedule	ra	always ...
Service	I;v FTP X I;v MYSQL X I;v RADIUS X +	
Action	☐ ACCEPT 0 DENY it LEARN	

Firewall/ Network Options

NAT	c)
IP Pool Configuration	UseOutgoing Interface Address Use Dynamic IP Pool
Preserve Source Port	CI

Security Profiles

AntNirus	c)	m default
Web Filter	CI	
DNSFilter	CI	
Application Control	CI	
IPS	CI	
SSLInspection		certifi c a t e - i n s p e c t i o n

Logging Options

Log Allowed Traffic	C>	Security Events
Generate Logs when Session Starts	C>	
Capture Packets	C>	

Comments, [Writeacomment](#)4, 0,1023

Enable this policy C>

ID7: Guests Internet

Purpose:

Allows devices in the Guests VLAN (Vlan 30) to access the internet through the WAN (port1) interface.

Explanation:

This firewall policy permits traffic originating from the Guests network to reach external destinations on the internet. The rule allows only the following essentials.

services:

- DNS – Required for resolving domain names (e.g., converting *google.com* to an IP address)
- HTTP – Standard web browsing
- HTTPS – Secure, encrypted web browsing
- PING – Allows basic connectivity testing.

Why it's important:

This policy provides controlled internet access to guest users while enforcing strict security inspection. Only necessary web-related services are allowed, minimizing exposure and preventing guests from using unauthorized or risky protocols. The applied security profiles help maintain network protection even for unmanaged guest devices.

Name0

Guests Internet

IncomingInterface

Guests(Vlan30)

Outgoing Interface

WAN (port1)

Source

all

+

Destination

all

+

Schedule

Co

always

Service

DNS

HTTP

HTTPS

PING

+

X

X

X

X

Action

HWfai0

DENY JS" LEARN

Firewall/ Network Options

NAT

0

IP Pool Configuration

UseOutgoing Interface Address

UseDynamic IP Pool

Preserve Source Port

C>

Security Profiles		
AntiVirus	0	-default ...
WebFilter	0	default } ...
DNSFilter	0	default } ...
Application Control	0	mdefault } ...
IPS	0	Bdefault ... ,
SSLInspection		EI certificate-inspection } ...

Logging Options

LogAllowedTraffic

0

Security Events

fffejfi@i.i..i

Generate LogswhenSessionStarts

0

Capture Packets

0

Comments |

Writeacomment...

AI

~1023

Enable thispolicy0



SNAT and DNAT

The session table output confirms that the FortiGate firewall is actively processing traffic across all VLANs. The presence of multiple ICMP sessions verifies successful connectivity testing between Staff, Guest, and Server networks, proving that inter-VLAN routing and firewall policies are functioning correctly. DHCP broadcast traffic (UDP ports 67/68) indicates that endpoint devices are successfully obtaining IP addresses from the FortiGate DHCP server. Additionally, the UDP/500 session confirms that IPsec VPN negotiation with the Branch network is active. Overall, the session list validates that the network is fully operational and that all configured policies and services are working as intended.

```
HQ-Firewall # get system session list
PROTO  EXPIRE  SOURCE          SOURCE-NAT      DESTINATION      DESTINATION-NAT
udp     178     0.0.0.0:68      -               255.255.255.255:67 -
icmp    42     192.168.40.1:49334 -             192.168.40.2:8   -
icmp    6      192.168.20.1:57420 -             192.168.20.3:8   -
icmp    23     192.168.30.1:41129 -             192.168.30.2:8   -
icmp    57     192.168.40.1:41151 -             192.168.40.4:8   -
icmp    17     192.168.20.1:49317 -             192.168.20.4:8   -
icmp    51     192.168.40.1:20667 -             192.168.40.3:8   -
icmp    36     192.168.30.1:28850 -             192.168.30.4:8   -
icmp    29     192.168.30.1:8366 -             192.168.30.3:8   -
icmp    0      192.168.20.1:16455 -             192.168.20.2:8   -
udp     60     192.168.72.100:500 -            192.168.72.200:500 -
```

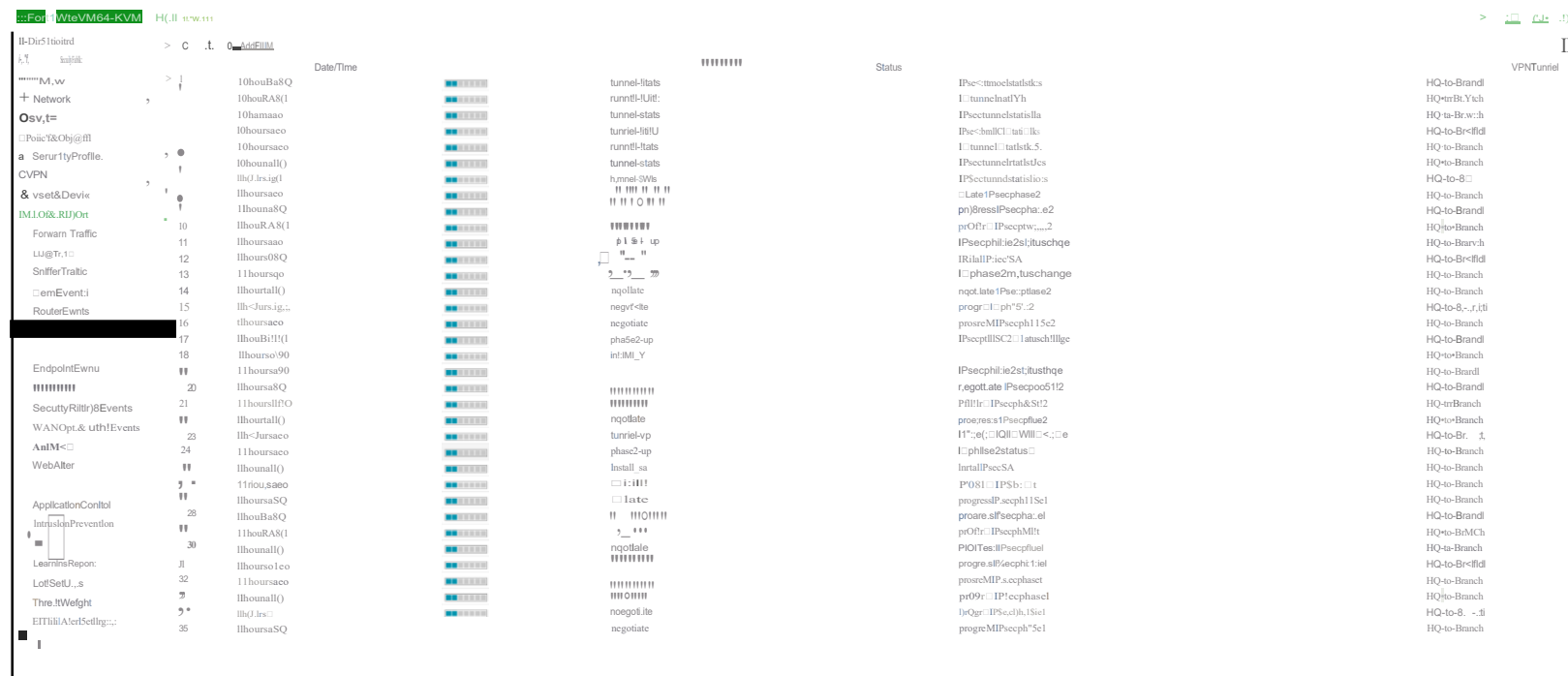
VPN

The firewall shows a connection between **192.168.72.100** and **192.168.72.200** using **UDP port 500**, which is the standard port for VPNs. This means these two devices are setting up or maintaining a **VPN tunnel** to securely communicate.

```
HQ-Firewall # get system session list
PROTO  EXPIRE  SOURCE          SOURCE-NAT      DESTINATION      DESTINATION-NAT
udp     28     0.0.0.0:68      -               255.255.255.255:67 -
icmp    53     192.167.20.2:21602 -             192.168.20.2:8   -
icmp    52     192.167.20.2:21346 -             192.168.20.2:8   -
icmp    51     192.167.20.2:21090 -             192.168.20.2:8   -
icmp    50     192.167.20.2:20834 -             192.168.20.2:8   -
udp     168     192.168.72.100:500 -            192.168.72.200:500 -

HQ-Firewall #
```

Output testing



Firewall

Refresh Reset Statistics Bring Up Bring Down

Name Type AemuteG.IT User Name Inwml Ob OutgoingData 1 PtlISe2select.or5

HQ-to-Branch Site to Site - FortGate 192.168.200 3.94 kB '6k8 HQ-to-Branch HQ-to-Branch

OAddFilter

	III	Date/Time	Source	Destination	Application Name	SecurityEWInts	Result
		10hour.sago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
> 2		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
> 3		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
9		10hoursago	cm oo:50:79:66:68:00	192.168.40.4	PING	V 84B/84B	StaffToServers(6)
10		10hoursago	191.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
11		10hoursago	191.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
12		10hours ago	191.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
13		10hoursago	192.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
14		10hoursago	191.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
15		10hoursago	191.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
16		10 hours ago	191.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
17		10hoursago	192.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
18		10hoursago	191.168.20.3	192.168.40.4	PING	V Accept:sessionstart	StaffToServers(6)
19		11hoursago	□_pq	192.168.20.3	PING	V 84B/84B	1
20		11hour.sago	cm PC2	192.168.20.3	PING	V 84B/84B	
21		11hoursago	l?_pq	192.168.20.3	PING	V 84B/84B	
22		11hour.sago	cmPC2	192.168.20.3	PING	V 84B/84B	
23		11hoursago	cmPC2	192.168.20.3	PING	V 84B/84B	
24		11 hours ago	cmPC2	192.168.20.3	PING	V 84B/84B	
25		11hoursago	cmPC2	192.168.20.3	PING	V 84B/84B	
26		11 hours ago	cmPC2	192.168.20.3	PING	V 84B/84B	
27		11hoursago	cmPC2	192.168.20.3	PING	V 84B/84B	
28		11hoursago	cmPC2	192.168.20.3	PING	V 84B/84B	
29		12 hours ago	cm PC2	192.168.10.3	PING	V 84B/112B	
30		12hoursago	cmPC2	192.168.10.3	ICMP/0/8	+	
31		12hoursago	cm PC2	192.168.10.3	PING	V 84B/112B	
32		12hoursago	cmPC2	192.168.10.3	ICMP/0/8	+	
33		12hoursago	cm PC2	192.168.10.3	PING	V 84B/112B	
34		12hoursago	cmPC2	192.168.10.3	ICMP/0/8	+	
35		12hoursago	cm PC2	192.168.10.3	PING	V 84B/112B	
36		12hoursago	cmPC2	192.168.10.3	ICMP/0/8	+	
37		12hoursago	cm PC2	192.168.10.3	PING	V 84B/112B	
38		12hoursago	a, PC2	192.168.10.3	ICMP/0/8	+	
39		12hoursago	cm PC2	192.168.10.3	PING	V 84B/112B	
40		12hoursago	cmPC2	192.168.10.3	ICMP/0/8	+	
41		12hoursago	cm PC2	192.168.10.3	PING	V 84B/112B	
42		12hoursago	□_pq	192.168.10.3	ICMP/0/8	+	
43		12hour.sago	cmPC2	192.168.10.3	PING	V 84B/112B	
44		12hoursago	cmPC2	192.168.10.3	ICMP/0/8	+	
45		12 hours ago	cmPC2	192.168.10.3	PING	V 84B/112B	
46		12hoursago	cmPC2	192.168.10.3	ICMP/0/8	+	
47		12 hours ago	cmPC2	192.168.10.3	PING	V 84B/112B	

The Guest2 client initially attempted DHCP discovery but did not receive an offer, which is common when the interface or DHCP service is still initializing. On the second attempt, the device successfully completed the full DORA process (Discover, Offer, Request, Acknowledge), receiving the IP address 192.168.30.2/24 from the DHCP server running on the FortiGate VLAN 30 interface (192.168.30.1). The client also received the correct gateway, DNS server, and a 7-day lease, confirming that the Guest VLAN DHCP configuration is working correctly and that Guest2 is fully integrated into the network.

```
Guest2> ip dhcp
DDD
Can't find dhcp server

Guest2> ip dhcp
DDORA IP 192.168.30.2/24 GW 192.168.30.1

Guest2> show ip

NAME           : Guest2[1]
IP/MASK        : 192.168.30.2/24
GATEWAY        : 192.168.30.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.30.1
DHCP LEASE     : 604781, 604800/302400/529200
MAC            : 00:50:79:66:68:05
LPORT         : 20078
RHOST:PORT     : 127.0.0.1:20079
MTU            : 1500
```

```
Guest2> ping 192.168.10.3
```

```
192.168.10.3 icmp_seq=1 timeout  
192.168.10.3 icmp_seq=2 timeout  
192.168.10.3 icmp_seq=3 timeout  
192.168.10.3 icmp_seq=4 timeout  
192.168.10.3 icmp_seq=5 timeout
```

```
Guest2> ping 192.168.72.200
```

```
192.168.72.200 icmp_seq=1 timeout  
192.168.72.200 icmp_seq=2 timeout  
192.168.72.200 icmp_seq=3 timeout  
192.168.72.200 icmp_seq=4 timeout  
192.168.72.200 icmp_seq=5 timeout
```

```
Guest2> ping 192.167.20.3
```

```
192.167.20.3 icmp_seq=1 timeout  
192.167.20.3 icmp_seq=2 timeout  
192.167.20.3 icmp_seq=3 timeout  
192.167.20.3 icmp_seq=4 timeout  
192.167.20.3 icmp_seq=5 timeout
```

```
Raduis> ip dhcp  
DDORA IP 192.168.40.2/24 GW 192.168.40.1
```

```
Raduis> show ip
```

```
NAME           : Raduis[1]  
IP/MASK         : 192.168.40.2/24  
GATEWAY         : 192.168.40.1  
DNS             : 8.8.8.8  
DHCP SERVER     : 192.168.40.1  
DHCP LEASE      : 604792, 604800/302400/529200  
MAC             : 00:50:79:66:68:08  
LPORT           : 20090  
RHOST:PORT      : 127.0.0.1:20091  
MTU             : 1500
```

```
Branch_Pc> ping 192.168.20.3
```

```
84 bytes from 192.168.20.3 icmp_seq=1 ttl=62 time=3.753 ms  
84 bytes from 192.168.20.3 icmp_seq=2 ttl=62 time=2.906 ms  
84 bytes from 192.168.20.3 icmp_seq=3 ttl=62 time=2.578 ms  
84 bytes from 192.168.20.3 icmp_seq=4 ttl=62 time=2.809 ms  
84 bytes from 192.168.20.3 icmp_seq=5 ttl=62 time=2.693 ms
```

```
Branch_Pc> █
```